

Graph Inequalities

Lesson 7-5

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*A horizontal line with marks at 0, 1, 2, 3, ...
showing where values fall*

Number line

A line with numbers in order. You use it to show answers.

*$x > 5$: open circle at 5 means 5 itself is NOT a
solution, but 5.1, 6, 7, ... are*

Open circle

An empty circle showing a number is NOT included.

*$x \geq 5$: closed circle at 5 means 5 IS a solution, and
so are 6, 7, 8, ...*

Closed circle

A filled-in circle showing a number IS included.

*For $x > 3$, the solution set includes 3.1, 4, 5, 10, 100
— any number greater than 3*

Solution set

All the numbers that make the inequality true.

Key Ideas & Notes

- Detective Okafor is building a timeline.
- The suspect was seen at the scene no earlier than 2:00 PM, meaning the arrival time t satisfies $t \geq 2$.
- She needs to plot this on a timeline to share with her team.
- How should she mark the number line?
- Graph each inequality on the number line. Decide whether to use an open or closed circle, then shade in the correct direction.

Think About It

- What does 'no earlier than 2:00 PM' mean about the time?
- Should 2:00 PM itself be included?
- Which direction on the number line would you shade?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Detective Okafor knows the suspect arrived 'no earlier than 2:00 PM,' so $t \geq 2$. On the timeline, should the mark at 2 be an open or closed circle, and which way should she shade?

Level 1 support

Start here: Because 2:00 itself counts, I use a closed circle at 2.

I use a ____ circle at 2 because 2:00 ____ included.

Uso un círculo ____ en 2 porque las 2:00 ____ incluidas.

I shade ____ because the times can be ____.

Sombreo ____ porque las horas pueden ser ____.

Word bank: number line open circle closed circle solution set

Listen for: Students choose a closed circle at 2 (because 'no earlier than' includes 2) and shade to the right since later times satisfy $t \geq 2$.

Level 2

How would the timeline look different if the clue were $t > 2$ instead of $t \geq 2$? Justify the change in the circle.

- For $t > 2$ I would use a ____ circle because ____.
- The boundary value 2 would ____ in that case.

EXPLORE

When you graphed $x > 4$ and $x \leq 7$, why did you use an open circle on one and a closed circle on the other?

Level 1 support

Start here: Open circles exclude the boundary; closed circles include it.

I used a ____ circle because the symbol ____ means the boundary is ____.

Usé un círculo ____ porque el símbolo ____ significa que el límite ____.

I shaded ____ because the solution set includes ____.

Sombree ____ porque el conjunto solución incluye ____.

Word bank: (open circle) (closed circle) (number line) (solution set)

Listen for: Students connect $>$ and $<$ to open circles (boundary not included) and $\geq$ and $\leq$ to closed circles, and shade toward the values in the solution set.

Level 2

Why does the graph of $x > 4$ represent infinitely many solutions even though it starts at one point? Explain using the solution set.

- The graph shows infinitely many solutions because ____.
- Every point in the shaded part is ____.

CONNECT

When would graphing an inequality show a real-life limit better than a single number?

Level 1 support

Start here: A graph shows every value that fits a rule, not just one.

A graph shows a real limit like ____ because ____.

Una gráfica muestra un límite real como ____ porque ____.

The ____ circle shows whether the limit value ____ allowed.

El círculo ____ muestra si el valor límite ____ permitido.

Word bank: (number line) (open circle) (closed circle) (solution set)

Listen for: Students name a real limit (minimum age, max weight, earliest time) and explain the circle type shows whether the boundary value is allowed.

Level 2

Critique: 'An open circle and a closed circle graph the same solution set.' Use $x > 5$ and $x \geq 5$ to show why this is wrong.

- This is wrong because $x \geq 5$ includes ____ but $x > 5$ ____.
- The two solution sets differ by ____.

Guided Examples

Example 1

Which graph represents $x < 5$?

Solution: $x < 5$ means 5 is NOT included (open circle) and values less than 5 are shaded (left).

Answer: A. Open circle at 5, shaded left

Example 2

Which graph represents $x \geq 3$?

Solution: $x \geq 3$ means 3 IS included (closed circle) and values greater than or equal to 3 are shaded (right).

Answer: A. Closed circle at 3, shaded right

Example 3

Which graph represents $x > 8$?

Solution: $x > 8$ means 8 is NOT included (open circle) and values greater than 8 are shaded to the right.

Answer: A. Open circle at 8, shaded right

Write About the Math

The Writing Revolution

I can explain my graph using the words number line, open circle, closed circle, and solution set.

1. Kernel Sentence subject + verb

Model: Number line is a line with numbers in order. You use it to show answers.

Recta numérica es una línea con números en orden. La usas para mostrar respuestas.

Write a kernel sentence about number line. Use a subject and a verb.

Escribe una oración base sobre recta numérica. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Number line matters in math

Recta numérica importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Number line matters in math because ____.

Recta numérica importa en matemáticas porque ____.

but
pero

Number line matters in math, but ____.

Recta numérica importa en matemáticas, pero ____.

so
entonces

Number line matters in math, so ____.

Recta numérica importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about number line.
Di un hecho verdadero sobre number line.

Number line ____.

Question *Pregunta*

Ask a question about number line.
Haz una pregunta sobre number line.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about number line.
Muestra entusiasmo sobre number line.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with number line.
Dile a un compañero qué hacer con number line.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I use a ____ **circle at 2 because 2:00** ____ **included.**

Uso un círculo ____ *en 2 porque las 2:00* ____ *incluidas.*

I shade ____ **because the times can be** ____.

Sombreo ____ *porque las horas pueden ser* ____.

I used a ____ **circle because the symbol** ____ **means the boundary is** ____.

Usé un círculo ____ *porque el símbolo* ____ *significa que el límite* ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which graph represents $x \leq 10$?

- A. Closed circle at 10, shaded left
- B. Open circle at 10, shaded left
- C. Closed circle at 10, shaded right
- D. Open circle at 10, shaded right

Show your work:

2. A number line shows a closed circle at 8 and is shaded to the left. Which inequality does this represent?

- A. $x \leq 8$
- B. $x < 8$
- C. $x \geq 8$
- D. $x > 8$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A theme park has two rides. Ride A requires height $h > 48$ inches. Ride B requires height $h \geq 48$ inches. A child is exactly 48 inches tall. Which ride can they go on?

Explain using number line graphs for both inequalities.

Sentence starter: For Ride A ($h > 48$), the graph has a ____ circle at 48. Since $48 > 48$ is ____, the child ____ ride A. For Ride B ($h \geq 48$), the graph has a ____ circle at 48. Since $48 \geq 48$ is ____, the child ____ ride B.

Show your work:

Reflect — Exit Ticket

How would you graph the inequality $x \geq 9$?

- A. Closed circle at 9, shaded right
- B. Open circle at 9, shaded right
- C. Closed circle at 9, shaded left
- D. Open circle at 9, shaded left

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. Closed circle at 10, shaded left — $x \leq 10$ means 10 IS included (closed circle) and values less than or equal to 10 are shaded (left).
2. **Try It 2:** A. $x \leq 8$ — Closed circle means the value is included ($\leq$ or $\geq$). Shaded left means less than or equal to: $x \leq 8$.
3. **Exit Ticket:** A. Closed circle at 9, shaded right — $x \geq 9$ means 9 is included (closed circle) and values are 9 or greater (shaded right).

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Number line is a line with numbers in order. You use it to show answers.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).